Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

ISWE406P – Agile Development Process and Devops Lab

Assignment III

Submitted By :

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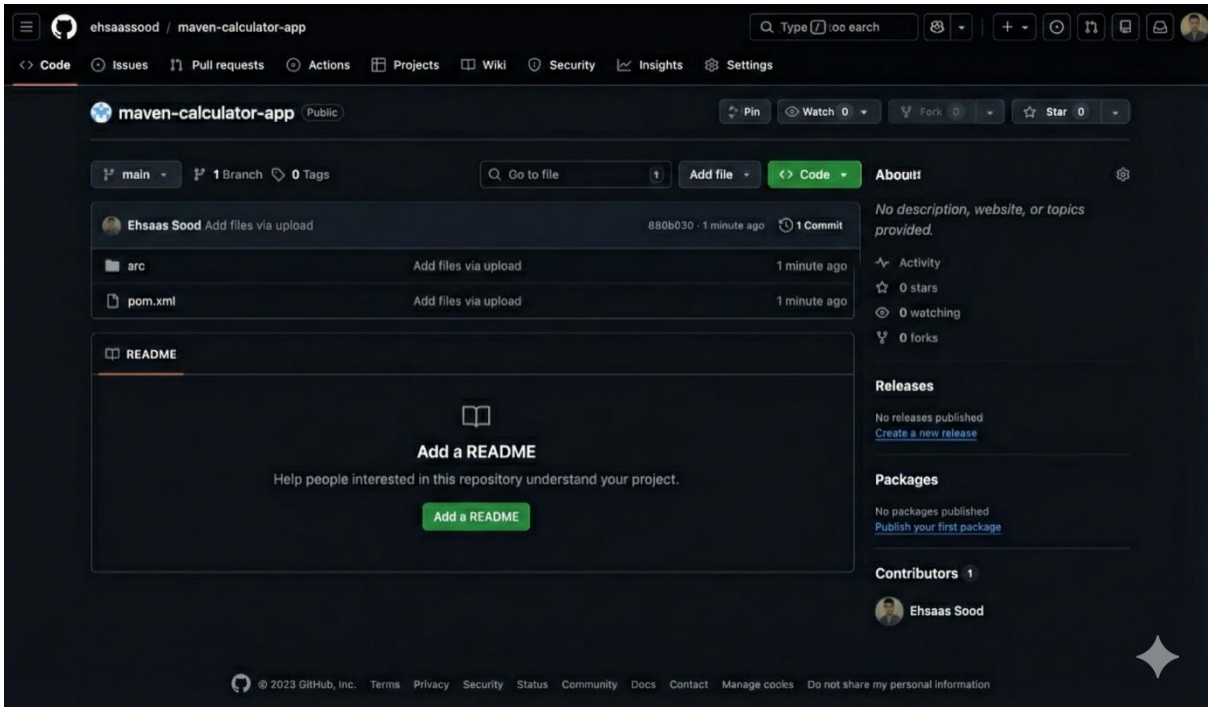
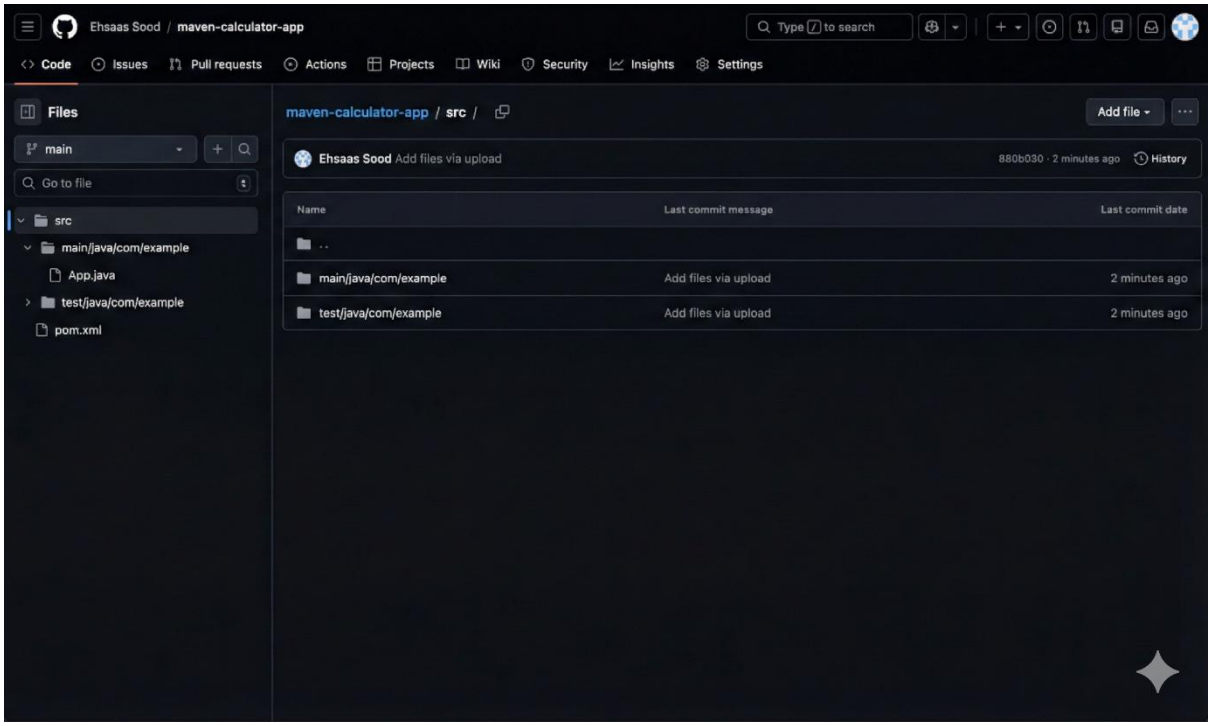
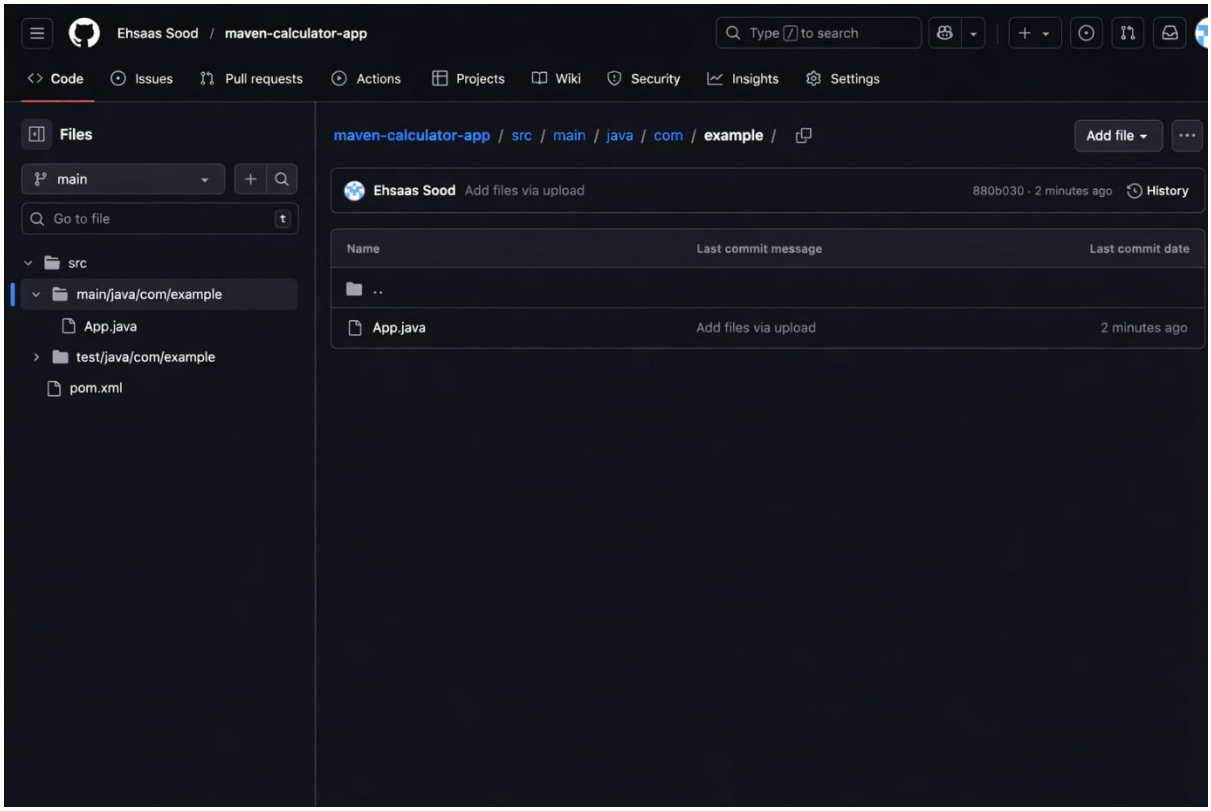
Register Number : 23MIS0074

Slot : L3 + L4

1. **Create a Maven Project in Git**
   - Create a new Git repository
   - Create a simple Maven project structure manually, including the following files and folders:
     - pom.xml
     - src/main/java/com/example/App.java
     - src/test/java/com/example/AppTest.java






Ehsaas Sood / maven-calculator-app
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Files
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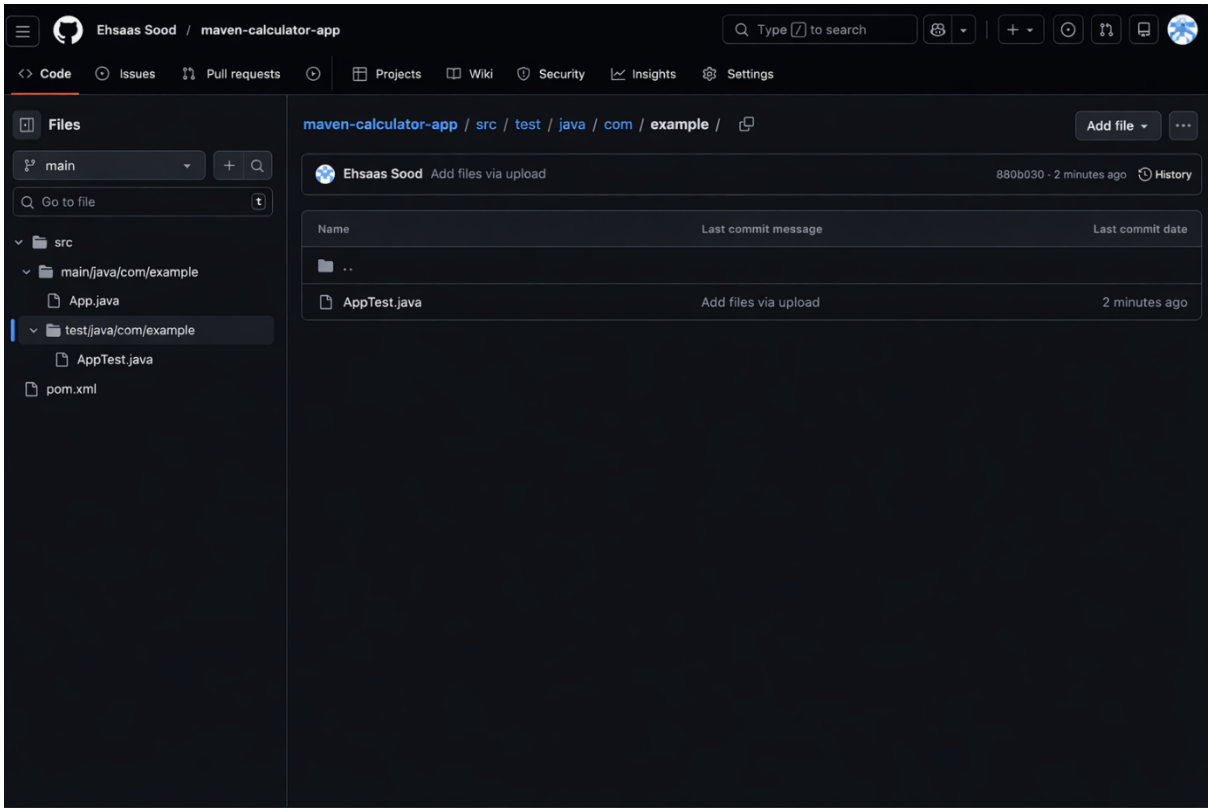
App.java
test/java/com/example
pom.xml
maven-calculator-app / src / main / java / com / example /
Add file
Ehsaas Sood Add files via upload
880b030 · 2 minutes ago
History
Name
Last commit message
Last commit date
..
App.java
Add files via upload
2 minutes ago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test/java/com/example
AppTest.java
pom.xml
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Add file
Ehsaas Sood Add files via upload
880b030 · 2 minutes ago
History
Name
Last commit message
Last commit date
..
AppTest.java
Add files via upload
2 minutes ago

2. Implement the Java Program

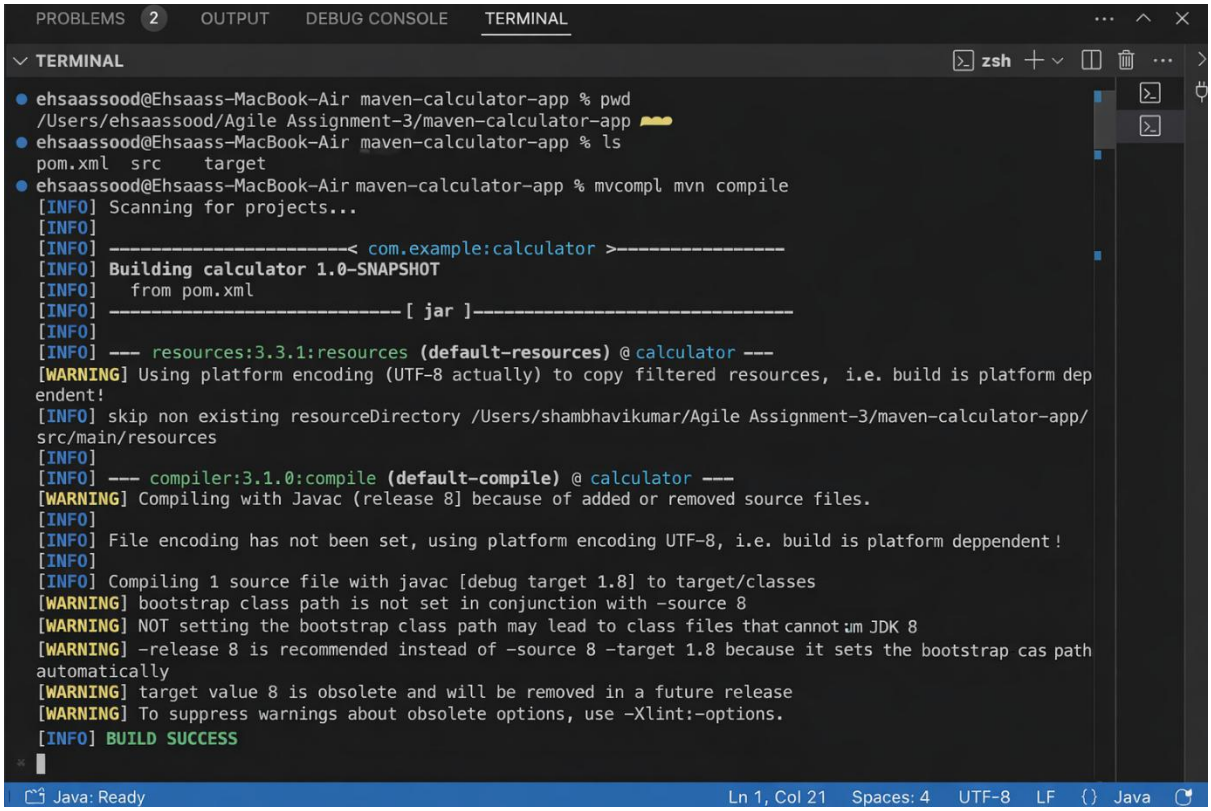
- Write a simple Java program in App.java under your newly created Git repository for any one of the following scenarios:
    - Student Grade Calculator
    - Bank Account Management System
    - Employee Salary Calculator
    - Library Book Management System
    - Simple Calculator Application
    - Write a corresponding unit test in AppTest.java.

3. Push and Clone the Repository

- Clone the repository using Visual Studio Code.
- Verify the project structure.

4. Build the Maven Project

- Use Maven commands in Visual Studio Code terminal to:
    - Compile the project
    - Run tests
    - Build the project (mvn clean install)



PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL

TERMINAL zsh

```
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.421 s
[INFO] Finished at: 2026-03-02T18:31:54+05:30
[INFO] ------------------------------------------------------------------------
Ehsasaaas Sood@Shambhavis-MacBook-Air maven-calculator-app % mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:calculator >-----------------------
[INFO] Building calculator 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ calculator ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platfor
m dependent!
[INFO] skip non existing resourceDirectory /Users/shambhavikumar/Agile Assignment-3/maven-calculator
-app/src/main/resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ calculator ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ calculator ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platfor
m dependent!
[INFO] skip non existing resourceDirectory /Users/shambhavikumar/Agile Assignment-3/maven-calculator
-app/src/test/resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ calculator ---
[INFO] Recompiling the module because of added or removed source files.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform depe
ndent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target/test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL

TERMINAL zsh

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.025 s -- in com.example.App
Test
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.763 s
[INFO] Finished at: 2026-03-02T18:32:02+05:30
[INFO] ------------------------------------------------------------------------
shambhavikumar@Shambhavis-MacBook-Air maven-calculator-app % mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:calculator >-----------------------
[INFO] Building calculator 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-
plugin/3.2.0/maven-clean-plugin-3.2.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-p
lugin/3.2.0/maven-clean-plugin-3.2.0.pom (5.3 kB at 15 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugin
s/35/maven-plugins-35.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins
/35/maven-plugins-35.pom (9.9 kB at 354 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/35/mave
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL

TERMINAL zsh

```
[INFO] Ehsaas Sood/ maven-calculator-app/ src /
[INFO] Finished at: 2026-03-02T18:31:54+05:30
[INFO] Scanning for projects...
[INFO] ------------------------------------------------------------------------
Ehsaas Sood: Add files via upload
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:calculator >-ven2/org/apache/maven/shared/maven-shared-u
[INFO] Building calculator 1.0-SNAPSHOT
[INFO]   from pom.xml           //repo.maven.apache.org/maven2/commons-io/commons-io/2.6/commons-io-2
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]                                                          fam-calculator
[INFO] --- resources:3.3.1:resources (default-resources) @ calculator ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platfor
m dependent!
[INFO] skip non existing resourceDirectory /Users/Ehambhavikumar/Agile Assignment-3/maven-calculator
-app/src/main/resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ calculator ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ calculator ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platfor
m dependent!
[INFO] skip non existing resourceDirectory /Users/Ehambhavikumar/Agile Assignment-3/maven-calculator
-app/src/test/resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ calculator ---
[INFO] Recompiling the module because of added or removed source files.
Ehsaas Sood@Shambhavis-MacBook-Air  maven-calculator-app % mvn test
[INFO] Deleting /Users/Ehsaas Sood/Agile Assignment-3/maven-calculator-app/target
[INFO] skip non existing resourceDirectory /Users/Ehsaas Sood/Agile Assignment-3/maven-calculator-a
pp/src/main/resources
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL

TERMINAL zsh

```
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ calculator ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform depe
ndent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target/test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
  not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class
path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ calculator ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.025 s -- in com.example.App
Test
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ calculator ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-manageme
nt/3.1.0/file-management-3.1.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-managemen
t/3.1.0/file-management-3.1.0.pom (4.5 kB at 104 kB/s)
```

Java: Ready Ln 1, Col 21 Spaces: 4 UTF-8 LF {} Java

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL

TERMINAL

```
[INFO] --- install:3.1.1:install (default-install) @ calculator ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
tinpver-util-19.19.18 maven-resolver-util-10.18.jam
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
anl-aps/3.1 18//maven-resolver-util-1.19.10.18.jam)
Downlloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
var-api/4.2.18/mmaven-resolver-util-1.19.10.18 hars
Downnloadeg from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
var-api24.128/mmaven-resolver-ap2-t1.19.18.17 jar)
Doownloadeg from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
lonrooo-6.4.19/mmaven-resolver-api-C1.19.10.18.jem)
Downlualing from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
ler-api/1.2.12 mmaven-resolver-ap2-t1.19.10.18.jer)
Downllading from central: htttp://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
ler-api/4.2.19/mmaven-resolver-ap2-C1.19.10.18.hars
Downlloding from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
lenrvoo/1.4.18/maven-resolver-ap2-t1.19.18.18 jar)
Downnloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
Uporeoe/2.4.22 mmaven-resolver-ap2 C1.19.10.18 jem)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-res
tooi-37-MaBBMO3/com/calchvaor-1j5 62 85.83.81 jam)
ce.imatot.jar 0rS:E-ISMTSnt/Oasoolator1m2/r8D09IIbv/com/example/calculator/1.0-SNAPSHOT/calculato
pomlexvIt-o /Users/ehes/ehsasssod/.m2/repository/com/example/aculator/1.0-SNAPSHOT/calculator-1.0-
SVOFCOOTlOrer-1.0-SNAPSHOT.jar
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
 Total time: 11.629 s
 Finished at: 2026-03-02T18:32:35+05:30
------------------------------------------------------------------------

ehsanssood@Ehsaaas-MacBook-Air maven-calculator-app %
```

Java: Ready Ln 1, Col 21 Spaces: 4 UTF-8 LF {} Java

maven-calculator-app

EXPLORER

MAVEN...
- src
  - main/java/com/ex...
    - App.java
  - test/java/com/ex...
    - AppTest.java
- pom.xml

Welcome

Visual Studio Code

Editing evolved

Start

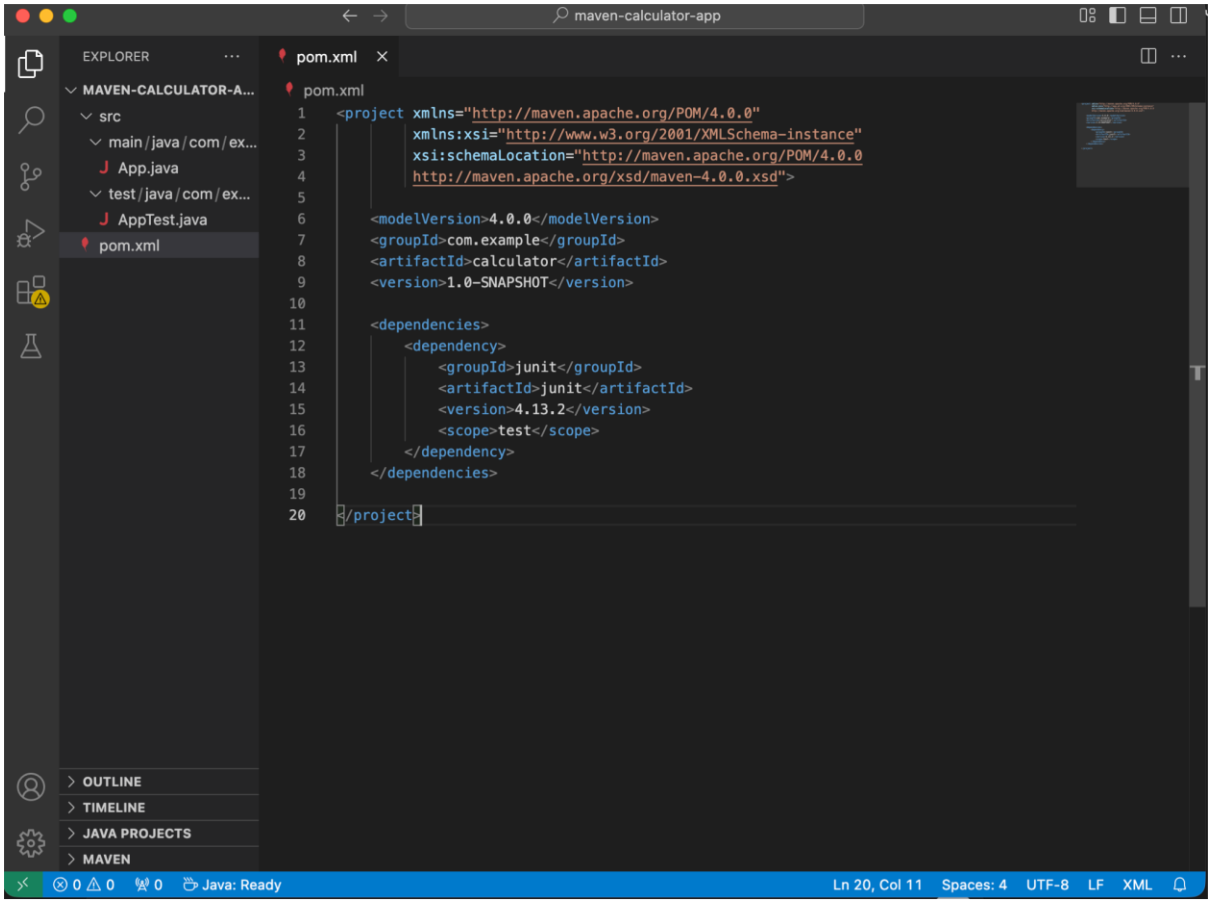
- New File...
- Open...
- Clone Git Repository...
- Connect to...

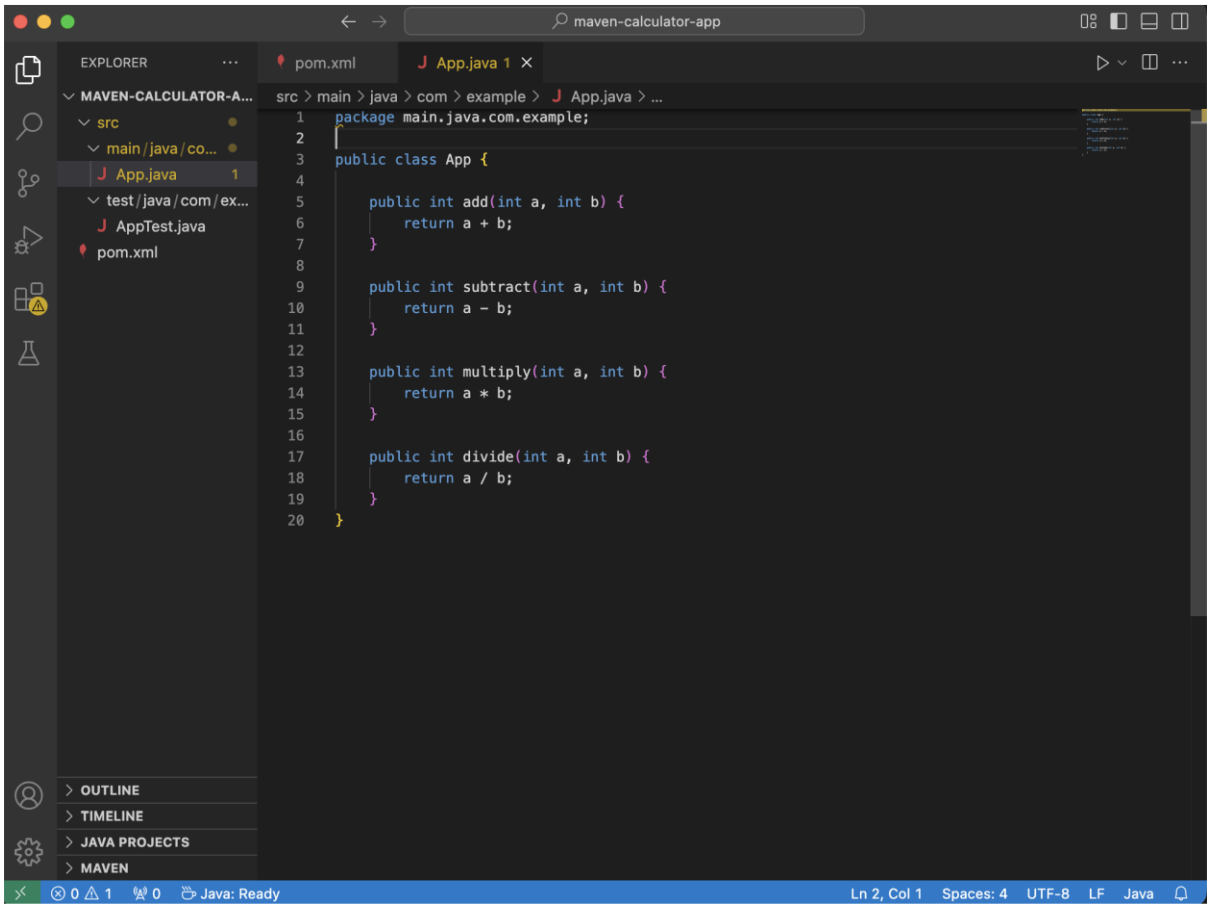
Walkthroughs

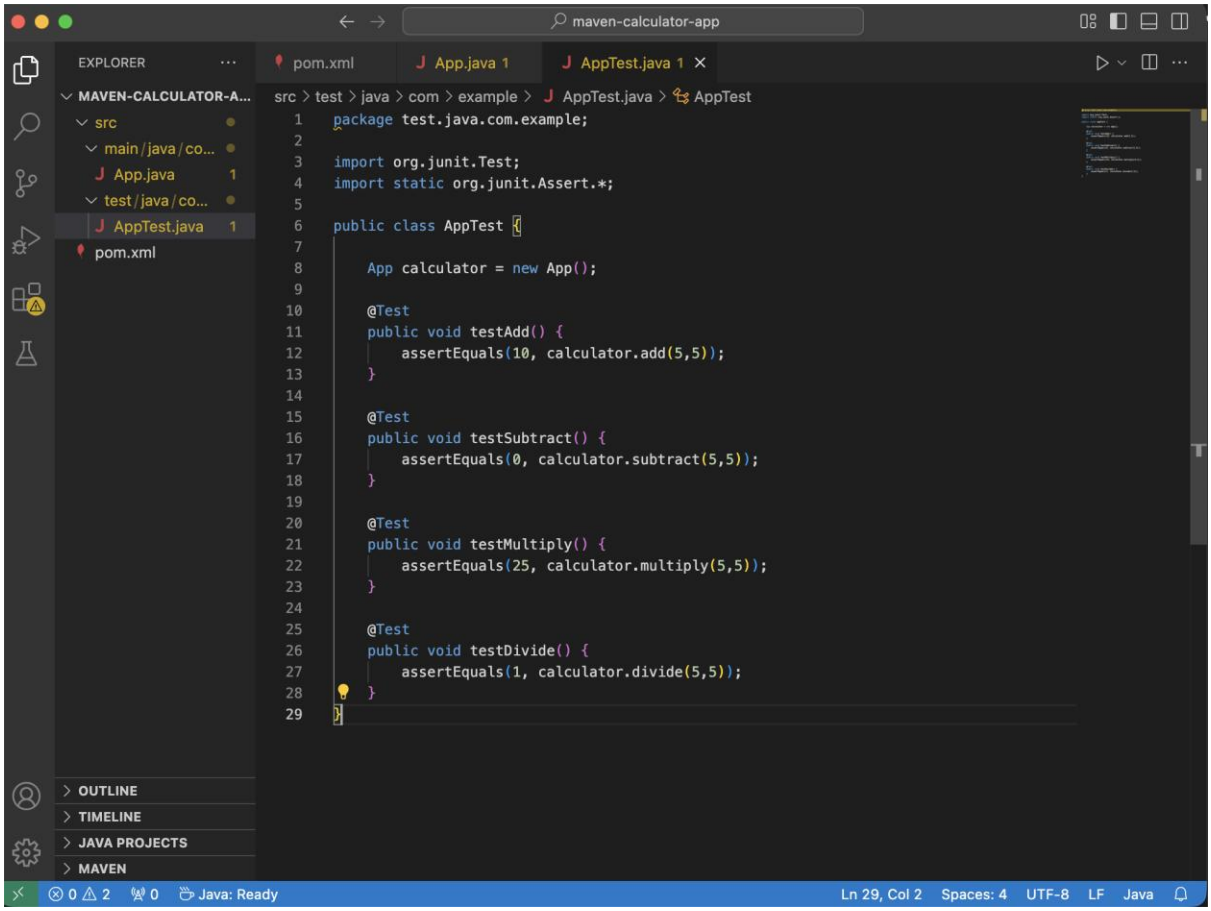
- Get Started with Pytho... Updated
- More...

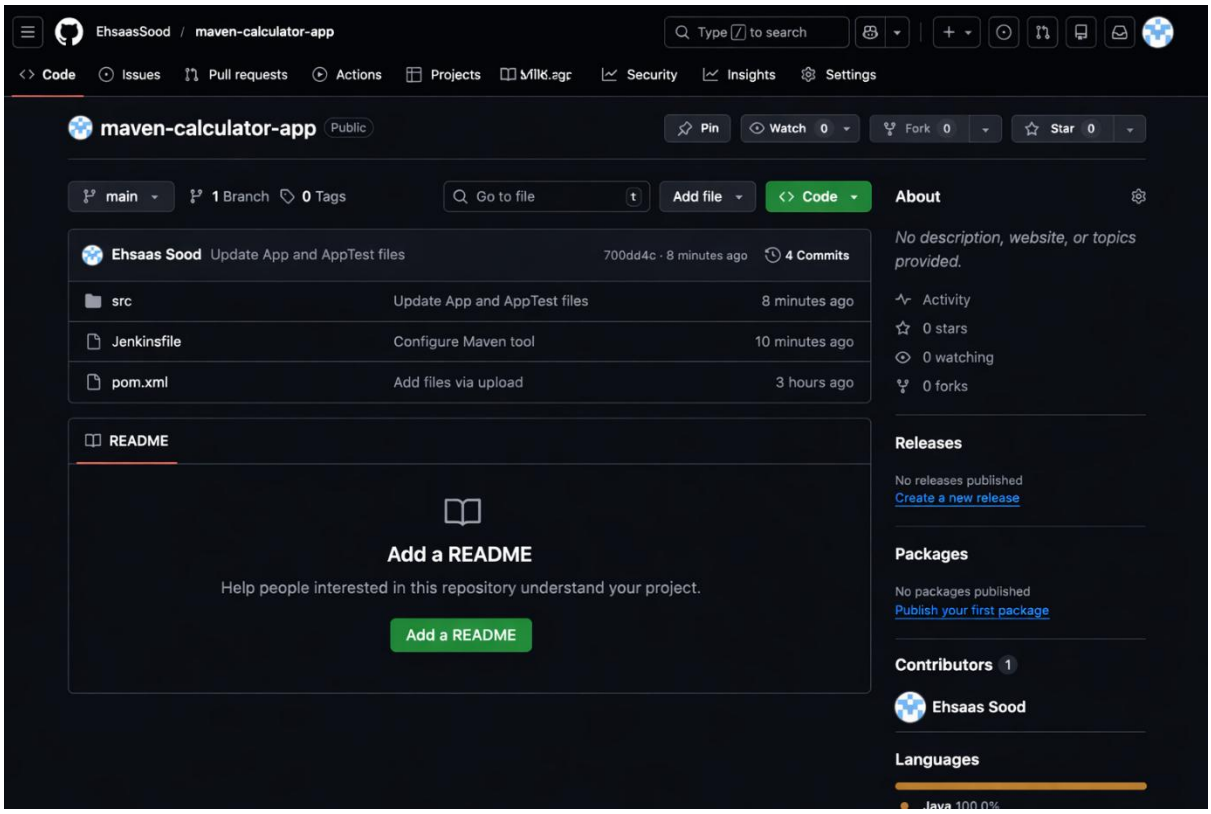
Recent

- Agile Development ~


maven-calculator-app
EXPLORER
MAVEN-CALCULATOR-A...
src
main / java / com / ex...
App.java
test / java / com / ex...
AppTest.java
pom.xml
pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>calculator</artifactId>
<version>1.0-SNAPSHOT</version>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>4.13.2</version>
<scope>test</scope>
</dependency>
</dependencies>
</project>
OUTLINE
TIMELINE
JAVA PROJECTS
MAVEN
Java: Ready
Ln 20, Col 11
Spaces: 4
UTF-8
LF
XML


maven-calculator-app
EXPLORER
MAVEN-CALCULATOR-A...
src
main / java / co...
App.java
test / java / com / ex...
AppTest.java
pom.xml
pom.xml
App.java 1
src > main > java > com > example > App.java > ...
package main.java.com.example;
public class App {
public int add(int a, int b) {
return a + b;
}
public int subtract(int a, int b) {
return a - b;
}
public int multiply(int a, int b) {
return a * b;
}
public int divide(int a, int b) {
return a / b;
}
}
OUTLINE
TIMELINE
JAVA PROJECTS
MAVEN
Java: Ready
Ln 2, Col 1
Spaces: 4
UTF-8
LF
Java

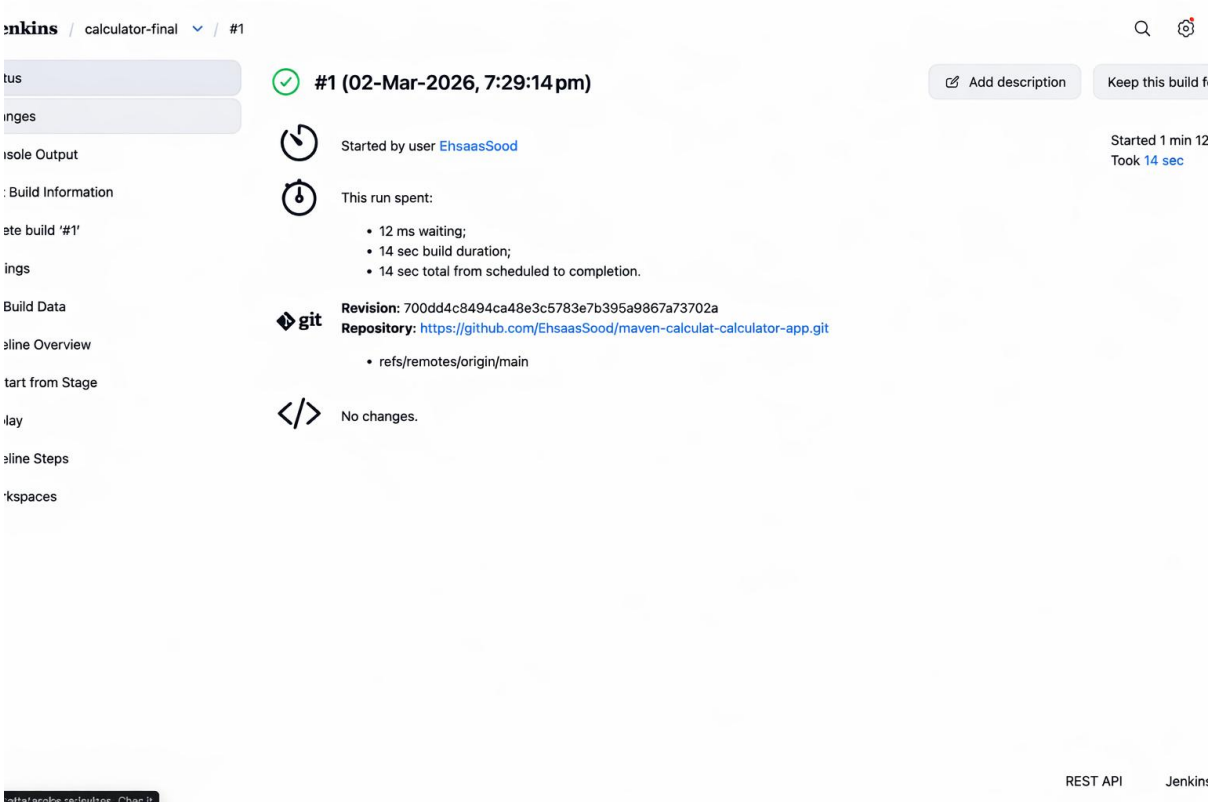


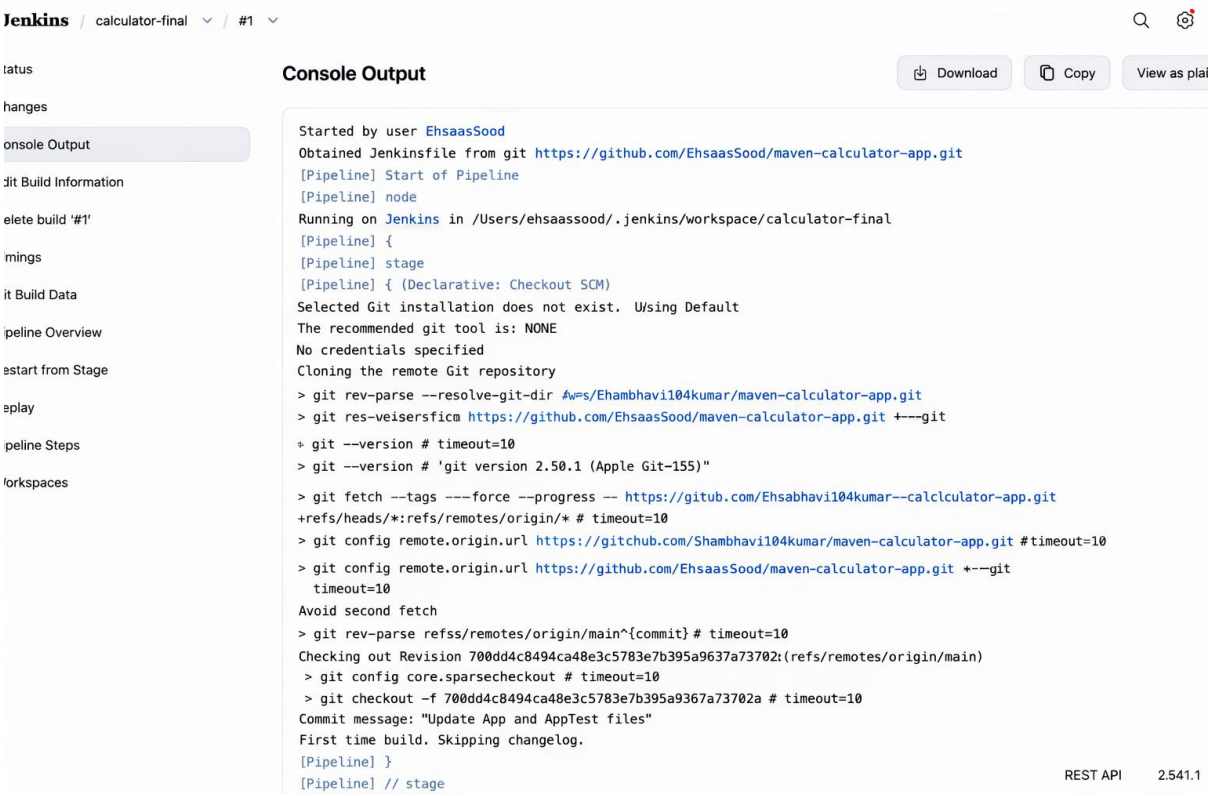


5. Create and Run Jenkins Pipeline

- Create a new Pipeline project in Jenkins.

- Create a pipeline script
- Execute the pipeline and verify successful build completion.





Jenkins / calculator-final / #1

```
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ calculator ---
[INFO] Building jar: /Users/ehsaassood/.jenkins/workspace/calculator-flarget/calculator-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- install:3.1.2:install (default-install) @ calculator ---
[INFO] Installing /Users/ehsaassood/.jenkins/workspace/calculator-final/pom.xml to
/Users/ehambhavikumar/.m2/repository/com/example/calculator/1.0-SNAPSHOTcalculator-1.0-SNAPSHOT.pom
[INFO] Installing /Users/ehsaassood/.jenkins/workspace/calculator-final/target/calculator-1.0-SNAPSHOT.jar to
/Users/ehambhavikumar/.m2/repository/com/example/calculator/1.0-SNAPSHOT/calculator-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.000 s
[INFO] Finished at: 2026-03-02T19:29:28+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] }
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.1